Idea 1: Do Sequential Tests Beat Max-Aggregation at a Matched False-Alarm Budget?

Matched-budget trajectory decision rules for AI-agent monitors:
rounds 1–4, with audit-corrected numbers

October 23, 2026 (supersedes the October 22 version)

Summary. AI-control evaluations turn per-step monitor scores into an audit decision by taking the *maximum* score and comparing it with an honest-run quantile. Over four rounds we compared max with accumulation rules (2nd/3rd-largest, top-k sums, mean, CUSUM, Shiryaev–Roberts, window GLR, conformal e-processes, OR multi-charts, rule choosers) at a *matched false-positive rate*, on 40 existing stream sets, 775 MonitoringBench attacks + 550 honest runs re-scored with and without history, persistent-state PR sequences re-scored by nine monitors with identical inputs, and all eight MOLE insider-threat corpora. Three independent audits re-derived every headline, and every threshold that matters below uses the conformal randomized rule (§4.3). Tags: [C] confirmatory (pre-specified before the analysed data were seen), [E] exploratory.

1. **Max remains the default.** Textbook sequential tests lose to max-type rules on almost every set [E]. No rule-selection procedure beats always-max out of sample (nested empirical chooser +0.001 [−0.006, +0.009] TPR at 1% FPR) [E].
2. **Cross-session accumulation does not generalize across monitor families.** Pooled over nine history-free per-PR monitors with identical inputs, top-3 sum − max at run-level FPR 0.05 is −0.066 [−0.147, +0.015] [C]. After Holm over the nine monitors only GPT-6 Luna's loss (−0.42) is significant. The earlier positive cases are fragile: GPT-4o (original scores) +0.134 [+0.024, +0.217] [E]; gpt-4.1-mini's +0.145 [+0.007, +0.230] appears only with its logprob-expected read-out (integer read-out of the same calls: −0.03) [E].
3. **Unreset streams: no demonstrated advantage for forgetting rules.** The round-3 CUSUM result for gpt-4.1-mini holds as computed (0.72 vs 0.59 at $\mathrm{ARL}_0 = 100$ PRs), but with CIs it is +0.132 [−0.086, +0.296], not significant [C]. Pooled over nine monitors no forgetting rule beats max at $\mathrm{ARL}_0 = 100$, and windowed top-3 sums lose (−0.116 [−0.178, −0.053]) [C]. Non-forgetting top-k sums collapse to ≈ 0.
4. **Insider threat (MOLE): no accumulation rule beats max** for a common monitor on any of 8 corpora (pooled single-day 3rd-max − max −0.064 [−0.151, +0.024]) [C]. A stronger day monitor (GPT-6 Luna) does not create a gain: +0.047 [−0.026, +0.125] with a joint account bootstrap [C]. Luna's account-level detection comes from attack days, not from an account-level signal.
5. **The proposed mechanisms are not supported.** "History double counting" is not supported once prompts are matched (the earlier effect was an instruction artifact) [C]. Repeat-spike informativeness does not predict the gain across monitors (cross-fitted Spearman +0.065 [−0.341, +0.460], 9 monitors; inconclusive) [C].
6. **Re-flagging monitor designs** (explicit re-raise rule, suspicion ledger, two samples) do not beat history-free whitened top-3 sum [E].

Spend: round 1 $109.56, round 2 $147.89, round 3 $161.36, round 4 $143.53, audit re-scoring $25.31 (OpenRouter); audit 3 and the threshold rerun $0.

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1 Motivation and question

A per-step monitor (an LLM judge) gives each agent action a suspicion score. Almost every AI-control paper and benchmark turns the resulting stream $s_1, \ldots, s_T$ into a trajectory decision the same way: audit if $\max_t s_t$ exceeds a threshold set at the $(1-\alpha)$ quantile of honest-run maxima, where α is the audit budget. BashArena, LinuxArena, MonitoringBench, Ctrl-Z, Control OSWorld and SHADE-Arena all do this.

Detection theory says max (a Shewhart chart) is optimal for a single large spike but weak when evidence is spread over many steps. For that case the textbook answer is a sequential statistic: CUSUM, Shiryaev–Roberts, GLR, e-processes. The question matters because:

- distributed attacks are an active concern;
- MonitoringBench reports that a 1% per-step FPR compounds to 6.7–17.4% per trajectory, and leaves a matched-FPR comparison of stepwise and full-trajectory monitoring to future work;
- three published attempts at simple accumulation failed: k-strikes rules cut safety from 61% to 44% (Redwood, *Retrying vs Resampling*); k-strikes failed in persistent-state control; Bayesian

score accumulation gave no gain (Meerkat).

Pre-registered mechanism hypothesis ("double counting"). Per-step monitors usually read the full history, so their scores are already cumulative judgments. Summing them double-counts evidence and inflates the honest tail of any accumulator. Accumulation should therefore help on *history-free* ("innovation") monitors and not on history-aware ones.

2 Study overview

Stage	Content	Code / outputs
Round 1 (Oct 20)	1A: 40 existing stream sets (LinuxArena 12 monitors, aksh-n, MRT, persistent state; $0). 1B: MonitoringBench 775 attacks + Honest-550 re-scored by gpt-4.1-mini with `full` vs `none` history, plus gpt-4.1-nano `none`. 1C: robustness (CUSUM k-grid, GLR window grid, conformal thresholds, Mondrian whitening).	`tracks/t1_matched_budget/`, `results/t1/`
Round 2 (Oct 21)	Persistent state re-scored by gpt-4.1-mini, Gemini 2.5 Flash, Qwen3-235B (all 691 runs, full diffs); MOLE GPT-5.3 and DeepSeek-V4 corpora; out-of-sample rule chooser over 67 sets; length-aware + top-k hybrid.	`tracks/r2_t1/`, `results/r2_t1/`
Audit 1 (Oct 21)	Corrected paired-gain bootstrap (C2); matched-prompt re-scoring (M4); nested chooser (M6/M7); agent-model confound (M9); realized FPR (m1); success-conditioning (m9).	`results/audit/`, `results/fixes/`
Round 3 (Oct 22)	A: all 8 MOLE corpora with one common history-free monitor, confound controls, horizon sweep. B: re-flagging monitor designs on persistent state and MonitoringBench. Long-horizon no-reset streaming.	`tracks/r3_t1/`, `results/r3_t1/`
Audit 2 (Oct 22)	Independent re-implementation of the round-3 analyses; 4 major, 10 minor issues (3 affect this report: A3, m5–m8).	`results/audit2/`
Round 4 (Oct 22)	Persistent state re-scored by six more monitor families (nine identical-input monitors); mechanism test across monitors; MOLE with GPT-6 Luna (and a gpt-4.1 subset); unreset streams with CIs.	`tracks/r4_cs/`, `results/r4_cs/`
Audit 3 (Oct 23)	Re-ran and independently re-implemented the round-4 analyses; 21 rows concern this report (read-out disclosure, Holm, MOLE account sharing, MOLE interpretation).	`results/audit3/`
Threshold fix (Oct 23)	Headline numbers re-run under the conformal randomized threshold rule (§4.3).	`results/fixes2/`

Table 1: Stages of Idea 1. All analyses ran on the laptop; all LLM scoring through OpenRouter.

3 Data

Confound controls (MonitoringBench). Prompts are action-only: agent reasoning is hidden from the monitor because its visibility differs between Honest-550 and the attacks. The final `submit` step is stripped from both classes (99% of honest runs end with it vs 78% of attacks). Argument names are normalized. Two confounds were found later and are handled in §5.2.1 and §5.2.2: the history modes differed in instruction wording, and the honest and attack sets differ in agent model.

Source	Honest / attack	Monitor(s)	Unit / onset label
aksh-n monitor-labels (7 settings)	564–1,396 / 564–1,305 per setting	Gemini-3.1-Pro streaming judge; *local* and *cumulative* scores, 0–10	step; ideal-flag step
LinuxArena monitor sweep	200 / 142	12 per-action monitors (gpt-4.1-nano … claude-opus-4-6), 0–100	step; harm step
ScaleAI MRT (SHADE / CUA-SHADE)	436 / 1,022	claude-3.5-sonnet chunk monitors ($N = 5, 7$; aware/unaware), 1–5	chunk of N messages
Persistent-state VibeCoding (round 1)	171 / 520	original GPT-4o per-PR monitors: `current_only`, `full_diffs`, `task_sequence`, trajectory	PR (5 per run)
Persistent state, re-scored (rounds 2–3)	155 / 120 gradual, 397 positional	gpt-4.1-mini, Gemini 2.5 Flash, Qwen3-235B-2507 (PR-review prompt, `current_only` vs `full_diffs`); round 3 adds gpt-4.1-mini design variants	PR; diffs re-read from `.eval` files
MonitoringBench + Honest-550	550 / 775	gpt-4.1-mini, `ctrl-basic-v1`, `full` vs `none`; gpt-4.1-nano `none`; audit: matched prompt `ctrl-matched-v1` on 200 / 175	step; `attack_step`
MOLE (HF `forgelab/mole`)	112–114 routine / 30–38 insider accounts per corpus	4 generators × single-day / multiday; Qwen3-235B account-day monitor (common), gpt-4.1-mini, MOLE z-score / peer-fit, activity-only	account-day over 30 workdays

Table 2: Score-stream sources. The 775 MonitoringBench attacks are all 475 S4 (distributed) attacks plus 100 each of S1–S3, stratified by side task (seed 20261020). The persistent-state paired filter keeps runs scored under every monitor.

Scoring. Per step, the logprob-expected 0–10 score from `gpt-4.1-mini` (token logprobs are returned via OpenRouter). `full`: task, all previous actions (truncated) and the current action. `none`: task and current action only. Passing `prompt_cache_key` and sending each trajectory's steps in order cut the projected full-run cost from $134 to about $101.[1]

4 Methods

4.1 Per-unit transforms and detectors

Each unit score s gets a randomized conformal p-value against honest calibration units, $p = (\#\{h_i > s\} + U(\#\{h_i = s\} + 1))/(n + 1)$, $U \sim \text{Unif}(0, 1)$. Sequential statistics use the whitened score $z = \Phi^{-1}(1 - p)$, $\approx \mathcal{N}(0, 1)$ on honest data (definitions in `docs/detectors.md`).

4.2 Matched-budget calibration

- **Honest-only thresholds.** 10 (round 1), 20 (rounds 2–3) or 40 (round 4) repeated 50/50 honest splits, grouped by main task when there are ≥ 30 tasks; 5-fold cross-fitting of the per-unit

[1] Source: `EXPERIMENT_LOG.md (2026-10-20 T1 entries)`.

Detector	Statistic (alarm when it exceeds h; trajectory stat = max over the running path)
max, k-th largest, top-k sum, mean	order statistics; top-k sums use whitened z unless stated
length-aware max	$-\log\big(1 - F(\max_{i\le t} s_i)^{\theta t^{\beta}}\big)$, analytic length-adjusted null
CUSUM(k)	$S_t = \max(0, S_{t-1} + z_t - k)$; $k = 0.5$ primary in round 1, $k = 1.5$ pre-specified in round 3
Shiryaev–Roberts(δ)	$\log R_t = \mathrm{logaddexp}(0, \log R_{t-1}) + \delta z_t - \delta^2/2$, $\delta = 1$
window GLR(W)	$G_t = \max_{m\le W} \max(0, \sum_{i=t-m+1}^{t} z_i)^2/(2m)$, $W = 20$ (plus grid)
conformal e-process	$\log E_t$, $E_t = \mathrm{mean}_\kappa \prod_{i\le t} \kappa p_i^{\kappa-1}$; SR e-detector $M_t = f(p_t)(1 + M_{t-1})$
OR multi-charts	max OR CUSUM (OR e-process), jointly calibrated
rule chooser	picks one of max / 2nd-max / top-3 sum / CUSUM($k \in [0.5, 2.5]$) from honest calibration data + n_{lab} labeled attacks

transforms. $\alpha \in \{0.05, 0.01, 0.003\}$ for trajectories; MOLE uses account-level FPR 0.10 and 0.05.

- **Realized FPR.** Under the plug-in rule (below) the threshold on $\approx$ 275 calibration runs over-alarmed at $\alpha = 0.01$: realized/target 0.77–2.30 (mean 1.28) on MonitoringBench.[2] Realized FPR is reported next to TPR where it matters.
- **Streaming.** Matched ARL by appending attacks to 0–4 honest test prefixes without reset; detection delay from the onset label; the non-restarting e-process is excluded from ARL comparisons.

4.3 Threshold convention (changed in this version)

Earlier versions set every threshold with the *plug-in* randomized quantile. For continuous scores its out-of-sample FPR is $(\lfloor \alpha n \rfloor + 1)/(n + 1)$, one order statistic too high when αn is an integer (1.98% at $n = 100$, $\alpha = 1\%$), and its randomization at ties helps only integer-score monitors.[3] **All thresholds now use the conformal randomized rule**: a new score s alarms with probability $\mathrm{clip}\big((\alpha(n+1) - \#\{S_i > s\})/(\#\{S_i = s\} + 1), 0, 1\big)$ over the n honest calibration scores S_i, which gives expected out-of-sample FPR exactly α for exchangeable scores, tied or continuous (`amd.evaluation.calibration.conformal_randomized_threshold`). The headline rows were re-run with only this switch changed.[4] For Idea 1 the change is small: the re-run headline point estimates moved by at most 0.012 (the descriptive FPR-0.01 persistent-state cell, whose thresholds sit at the calibration maximum, by 0.03), and no verdict changed except that gpt-4.1-mini's persistent-state gain now has an iid interval that excludes 0 (§6.1). Numbers quoted below are conformal when the row was re-run; tables and cells that were not re-run keep their plug-in values and say so. Unreset-stream thresholds are set by ARL matching and are not affected.

4.4 Confirmatory and exploratory results

[C] marks an analysis whose comparison and criterion were fixed before the analysed data were seen: the round-1 history DiD, the round-3 pre-specified comparisons (3rd-max, top-3 sum, CUSUM k=1.5 and the chooser vs max), and the round-4 analyses whose scripts carry their pre-specification

[2] Source: `results/fixes/t1/m1_realized_fpr.csv`.
[3] Sources: `results/audit3/corrections.csv`; `results/audit3/wb_convention.csv`.
[4] Source: `results/fixes2/headline_threshold_rerun.csv`.

(persistent-state pooled top-3 − max at FPR 0.05, mechanism test, MOLE pooled 3rd-max − max at account FPR 0.10, unreset streams P3a/P3b). Audit 3 found that two of those round-4 scripts (`mechanism.py`, `mole_r4_analysis.py`) were committed after the pre-commit, but session transcripts show they were written before any of their outputs existed. Everything else, including every robustness check and every audit re-analysis, is [E].

4.5 Inference (corrected after audit 1)

The round-1 and round-2 gain CIs drew new attack-bootstrap indices in each split and averaged replicates across splits, which shrinks the interval by about $\sqrt{n_{\text{splits}}}$; they also held thresholds fixed and resampled attacks i.i.d. (audit issue C2).[5] All gains in this report use the corrected paired bootstrap (`amd.evaluation.paired`): one joint resample per replicate of honest runs (clustered by task or account, thresholds recomputed in every split) and attacks (same clusters), shared across splits and across the terms of each contrast. Corrected CIs are 6.4× wider (median); 792 of 1,560 round-1 gains remain significant (previously 1,397).[6] Persistent-state results also report a 5-app cluster sensitivity interval. Rule choosers use *nested* calibration (select on one calibration half, threshold on the other) and a task-grouped labeled draw; set-level correlations get a cluster bootstrap over 14 set families (MOLE: over corpora).

5 Round 1: existing streams and controlled re-scoring

5.1 1A: 40 existing stream sets ($0) [E]

Set	max	2nd	top-3	CUSUM	SR	GLR	e-proc	len-max
LinuxArena: claude-opus-4-6	**0.95**	0.88	0.92	0.66	0.65	0.69	0.81	**0.97**
LinuxArena: gemini-3.1-pro	0.93	0.85	0.88	0.56	0.56	0.70	0.68	**0.94**
LinuxArena: gpt-5.4	0.77	0.72	0.71	0.32	0.32	0.56	0.49	**0.84**
LinuxArena: gpt-5-mini	0.69	0.56	0.60	0.37	0.36	0.44	0.47	**0.70**
LinuxArena: glm-5	0.84	0.78	**0.89**	0.64	0.63	0.70	0.79	0.88
LinuxArena: gpt-4.1-nano	0.20	0.13	0.14	0.13	0.12	0.13	0.16	**0.24**
aksh-n BashArena sonnet45 (cum.)	0.98	0.98	0.96	0.31	0.32	0.68	0.76	**0.99**
aksh-n shade (local)	0.82	0.82	0.84	0.62	0.62	0.64	0.45	0.81
MRT sonnet-3.5 N=5	0.21	0.22	0.18	0.02	0.01	0.05	0.03	**0.58**
MRT sonnet-3.5 N=5 superaware	0.33	0.37	0.26	0.02	0.02	0.04	0.06	**0.83**
PR `current_only`: gradual	0.16	0.24	0.31	0.28	**0.33**	0.21	0.15	0.16
PR `current_only`: positional	0.19	0.16	**0.30**	0.26	0.29	0.24	0.24	0.17
PR `full_diffs`: gradual	**0.24**	0.09	0.16	0.22	0.22	**0.25**	0.23	0.23
PR trajectory monitor: gradual	0.35	0.17	0.31	**0.37**	**0.37**	**0.37**	0.17	0.33

Table 3: TPR at matched trajectory FPR $\alpha = 0.05$ (point estimates; selected sets). CUSUM is $k = 0.5$. Source: `results/t1/tables/r1_tpr.md` (all 40 sets and $\alpha \in \{0.05, 0.01, 0.003\}$). The audit changed none of these point estimates, only the gain CIs. Plug-in thresholds (not re-run).

[5] Source: `results/audit/issues.csv (C2)`.
[6] Source: `results/fixes/corrections.csv (C2, row ''t1 r1 (40 stream sets)'')`.

- **Max-type rules dominate.** At $\alpha = 0.01$, max, 2nd-max or length-aware max matched or beat every accumulator in 86% of aksh-n sets, 67% of LinuxArena monitors and 83% of MRT sets. Median TPR(CUSUM k=0.5) − TPR(max): −0.39, −0.25, −0.01. These point-estimate rankings are unchanged by the audit.[7]
- **Accumulation gain falls with honest autocorrelation** (Fig. 1): Spearman −0.57 ($\alpha = 0.05$) and −0.45 ($\alpha = 0.01$) between the CUSUM−max gain and the honest lag-1 autocorrelation of whitened scores. These correlations treat sets as independent and are descriptive (see §6.4).
- **Timing.** On aksh-n, max alarms at the ideal flag step (median delay 0) and CUSUM 3–25 steps later. The onset is the adjudicator's `ideal_flagging_step`, itself an LLM judgment, so a delay of 0 for a spike rule is partly by construction.[8]
- **Length-aware max** wins where honest and attack lengths differ. MRT N=5: +0.365 [+0.127, +0.445]; superaware: +0.506 [+0.344, +0.549] (corrected CIs). An attacker who pads benign steps controls length, so the gain is conditional on unadapted attack lengths.[9]
- **Persistent state (original GPT-4o PR monitors, gradual attacks, $\alpha = 0.05$).** History-free `current_only`: top-3 sum − max = +0.134 [+0.024, +0.217], $p(\leq 0) = 0.008$; 5-app cluster sensitivity [+0.038, +0.237] (**holds**; plug-in +0.146 [+0.025, +0.226]).[10] Realized FPR is above nominal under both rules (max 0.059, top-3 0.066; $n_{\text{cal}} = 84$). Plug-in only (not re-run): CUSUM k=0.5 − max = +0.116 [+0.006, +0.171], app-cluster [−0.006, +0.191] (**weakened**); history-aware `full_diffs` top-3 sum − max = −0.074 [−0.143, −0.003] (**holds**).[11] Round 4 puts this gain in context: it does not generalize across monitor families (§8).

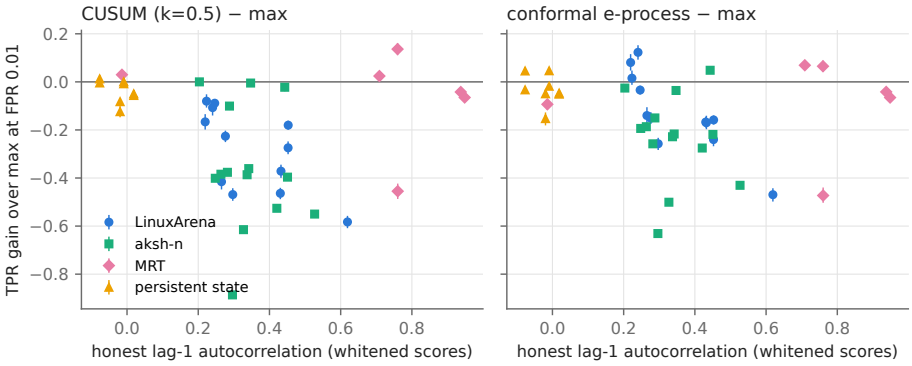


Figure 1: Round 1: gain over max (TPR at FPR 0.01; point estimates) vs honest lag-1 autocorrelation of whitened scores, one point per stream set. Sets with strongly autocorrelated honest scores lose the most.

5.2 1B: controlled history experiment on MonitoringBench [E]; pre-registered DiD [C]

Same monitor (`gpt-4.1-mini`); the history mode differs (`full` vs `none`). Table 4 gives TPR with realized FPR.

[7] Source: `results/fixes/corrections.csv (C2, 40 stream sets)`.
[8] Source: `results/fixes/corrections.csv (m12)`.
[9] Source: `results/fixes/corrections.csv (m7)`.
[10] Source: `results/fixes2/headline_threshold_rerun.csv`.
[11] Source: `results/fixes/t1/c2_headline.md`.

	$\alpha = 0.05$				$\alpha = 0.01$			
Detector	full	(FPR)	none	(FPR)	full	(FPR)	none	(FPR)
max	0.226	0.054	**0.322**	0.047	0.171	0.019	0.156	0.014
2nd-max	**0.342**	0.044	0.276	0.047	0.190	0.008	0.152	0.008
top-3 sum	0.324	0.060	0.304	0.051	**0.195**	0.008	**0.190**	0.009
CUSUM k=0.5	0.073	0.054	0.087	0.057	0.030	0.013	0.039	0.023
e-process (power)	0.130	0.049	0.127	0.048	0.052	0.011	0.065	0.016
length-aware max	0.215	0.053	0.279	0.045	0.153	0.015	0.124	0.009
OR(max+CUSUM)	0.201	0.062	0.247	0.059	0.137	0.014	0.086	0.013

Table 4: TPR at matched trajectory FPR with realized held-out FPR (plug-in thresholds, not re-run), 775 attacks vs 550 honest, original prompts (`ctrl-basic-v1`). Source: `results/fixes/t1/m1_realized_fpr.csv`; full detector list in `results/t1/tables/r2_main.md`.

Corrected gains (original prompts). [12] With history (`full`), 2nd-max and top-3 sum beat max at $\alpha = 0.05$ (+0.116 [+0.026, +0.183]; +0.098 [+0.042, +0.156]), but not at $\alpha = 0.01$ (top-3 sum +0.024 [−0.025, +0.085], **dropped** relative to the earlier [0.018, 0.031]). Without history, no accumulator beats max at any α. CUSUM k=0.5, the e-process, length-aware max and the OR chart lose to max in every cell (e.g. CUSUM full, $\alpha = 0.01$: −0.141 [−0.175, −0.101]). The weak history-free monitor (`gpt-4.1-nano`) is the exception: top-3 sum 0.082 vs max 0.056 at $\alpha = 0.01$, gain +0.026 [+0.001, +0.058] (**holds**, barely).

The pre-registered test (original prompts). Difference-in-differences (none − full) of the gain over max: CUSUM at $\alpha = 0.01$ +0.024 [−0.050, +0.072] (all) and +0.029 [−0.048, +0.090] (S4), both not significant. At $\alpha = 0.05$ the DiD is significantly *negative* for 2nd-max (−0.162 [−0.236, −0.056]), top-3 sum (−0.116 [−0.184, −0.041]) and CUSUM (−0.083 [−0.161, −0.004]).[13] That looked like "accumulators gain more with history", but the next subsection shows it was produced by the prompts.

5.2.1 Matched prompts remove the history effect (audit M4)

The `full` prompt told the monitor that continuing or completing earlier suspicious activity "counts as evidence"; the `none` prompt told it to score low even if an action could be part of a larger plan. The re-flagging mechanism proposed in round 1 is exactly what the `full` instruction asks for. We re-scored 200 Honest-550 runs and 175 attacks (100 S4 + 25 each of S1–S3) with `ctrl-matched-v1`: identical instructions, only the previous-actions block differs (21,786 calls, $25.31).

- On the same subset the original prompts reproduce the negative DiD at $\alpha = 0.05$; with matched prompts it vanishes. "Accumulators gain more with history" was an **instruction artifact** (**dropped**).
- "Gains concentrate in history-free mode" is still **not supported**: no matched DiD is significantly positive.
- History-free max is *lower*, not higher, with matched prompts: none − full max TPR at $\alpha = 0.05$ is −0.091 [−0.186, +0.038] vs +0.089 [−0.014, +0.159] with the original prompts (**reversed**). The original `none` instruction helped the history-free monitor.[14]

[12] Source: `results/fixes/t1/gains_r2mb_corrected.csv`.
[13] Source: `results/fixes/t1/did_r2mb_corrected.csv`.
[14] Source: `results/fixes/corrections.csv (M4)`.

α	Detector	DiD, original prompts	DiD, matched prompts	instruction effect (orig − matched)
0.05	2nd-max	−0.153 [−0.245, −0.019]	−0.023 [−0.180, +0.090]	−0.130 [−0.250, +0.064]
0.05	top-3 sum	−0.122 [−0.196, +0.003]	+0.030 [−0.110, +0.123]	−0.152 [−0.245, +0.035]
0.05	CUSUM k=0.5	−0.091 [−0.163, +0.029]	+0.089 [−0.040, +0.201]	−0.180 [−0.289, −0.025]
0.01	2nd-max	−0.001 [−0.194, +0.086]	−0.090 [−0.210, +0.037]	+0.089 [−0.107, +0.173]
0.01	top-3 sum	+0.063 [−0.119, +0.116]	−0.082 [−0.164, +0.049]	+0.145 [−0.063, +0.190]
0.01	CUSUM k=0.5	+0.021 [−0.130, +0.073]	+0.068 [−0.017, +0.149]	−0.047 [−0.201, +0.014]

Table 5: DiD (none − full) of the gain over max on the 200 / 175 subset, all attacks (plug-in thresholds). Positive = accumulation gains concentrate in history-free mode. Source: `results/fixes/t1/m4_tables.md`.

- Re-flagging: share of attacks with ≥ 2 steps above the honest q95, full − none: all +0.03 [−0.03, +0.09], S1 +0.08 [−0.08, +0.24] (matched; **weakened**).[15]

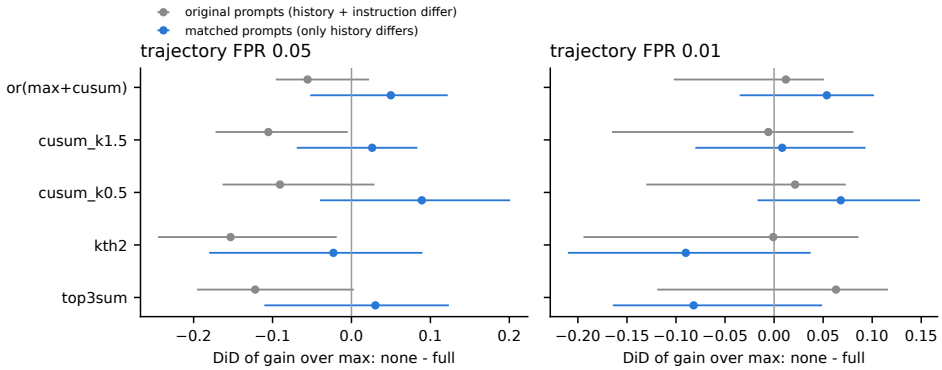


Figure 2: DiD (none − full) of the gain over max, original prompts (grey; history + instruction differ) vs matched prompts (blue; only history differs), with corrected joint task-cluster bootstrap CIs.

5.2.2 Agent-model confound in absolute TPR (audit M9)

Honest-550 is 100% claude-opus-4-5, while 281 of the 775 attacks (36%) are claude-opus-4-6. Max TPR at $\alpha = 0.01$: opus-4-5 attacks 0.121 [0.094, 0.165] (full) / 0.117 [0.083, 0.210] (none); opus-4-6 0.258 [0.200, 0.316] / 0.224 [0.163, 0.325]. Like-for-like detection is therefore about 12%, not "17–20%". Top-3 sum on opus-4-5 attacks: 0.161 [0.093, 0.241] (full), 0.148 [0.081, 0.218] (none). Detector comparisons are paired and less affected.[16]

5.3 1C: robustness ($0) [E]

- **k is critical.** Small-k CUSUM accumulates honest noise on whitened LLM scores and also over-alarms (realized FPR 0.020–0.032 at $\alpha = 0.01$). $k \approx 1.5$–2 turns CUSUM into a smoothed max and recovers max-level TPR. Part of the poor showing of CUSUM k=0.5, SR and long-window GLR is mis-tuning.

[15] Source: `results/fixes/t1/m4_mechanism_repeat_share.csv`.
[16] Source: `results/fixes/t1/m9_tpr_by_agent_model.csv`.

Detector ($\alpha = 0.01$)	TPR (full)	realized FPR	TPR (none)	realized FPR
max	0.164	0.010	0.146	0.013
CUSUM $k = 0.25$	0.024	0.020	0.031	0.028
CUSUM $k = 0.5$	0.032	0.020	0.043	0.032
CUSUM $k = 1$	0.097	0.026	0.070	0.027
CUSUM $k = 1.5$	**0.186**	0.009	0.155	0.015
CUSUM $k = 2$	0.162	0.006	**0.159**	0.014
top-2 / top-3 / top-5 sum	0.185 / 0.205 / 0.169	0.011–0.013	0.198 / 0.186 / 0.149	0.009–0.016
GLR $W = 5/10/20$	0.144 / 0.083 / 0.058	0.021–0.025	0.088 / 0.063 / 0.057	0.017–0.030

Table 6: MonitoringBench parameter grid (775 attacks, original prompts; plug-in thresholds). Source: `results/t1/round3/robust/matched_kgrid_\{full,none\}.csv`.

- **Dependence breaks i.i.d. guarantees.** The Ville ($1/\alpha$) threshold of the e-process over-alarms at about 2× nominal on dependent honest streams; the analytic length-aware threshold holds nominal FPR.[17]
- Conformal thresholds give the same ranking; keeping the submit step changes nothing; Mondrian per-task whitening inflates FPR under grouped splits (inconclusive).

6 Round 2: cross-session accumulation across monitors, MOLE, rule chooser [E]

6.1 Persistent state re-scored by four monitors

All 691 runs were re-scored from the full diffs (the processed data had kept 3k of a median 39k characters) with the paper's PR-review prompt in two modes that share an identical system prompt and differ only by the previous-diffs block, so the M4 confound does not apply here.[18] Paired sets: 155 honest / 120 gradual / 397 positional runs, 20 splits.

History-free monitor	max	top-3	top-3 − max [95% CI]	$p(\leq 0)$	app-cluster	DiD (free − aware)	verdict
gpt-4.1-mini*	0.58	0.73	+0.145 [+0.007, +0.230]	0.018	[−0.005, +0.249]	−0.169 [−0.368, −0.019]	weakened
GPT-4o (original scores)	0.15	0.26	+0.101 [+0.015, +0.208]	0.015	[+0.005, +0.228]	+0.173 [+0.055, +0.271]	holds
Gemini 2.5 Flash	0.23	0.25	+0.021 [−0.068, +0.101]	0.346	[−0.077, +0.139]	−0.028 [−0.182, +0.091]	dropped
Qwen3-235B-2507	0.44	0.42	−0.025 [−0.121, +0.061]	0.724	[−0.133, +0.116]	+0.127 [−0.040, +0.244]	dropped

Table 7: Gradual (multi-PR) attacks, run-level FPR 0.05. TPRs: `results/r2_t1/ps_tables.md`; corrected gains and DiDs: `results/fixes/r2_t1/ps_gains_corrected.csv`, `results/fixes/r2_t1/ps_did_corrected.csv`, `results/fixes/corrections.csv`. *The gpt-4.1-mini gain is the conformal-rule re-run (plug-in: +0.145 [−0.001, +0.228], $p(\leq 0) = 0.026$, app-cluster [−0.016, +0.246]; `results/fixes2/headline_threshold_rerun.csv`); it uses the logprob-expected read-out (§8.1). All other cells, and the DiDs, are plug-in. The GPT-4o row uses the paired filter (153 honest), hence +0.101 rather than the round-1 +0.134 (conformal).

- **The history-free gain is monitor-specific.** It holds for GPT-4o, is borderline for gpt-4.1-mini (iid interval excludes 0 only under the conformal rule, $p(\leq 0) = 0.018$; the app-cluster

[17] Source: `EXPERIMENT_LOG.md (2026-10-20 T1 results, Round 3)`.
[18] Source: `results/fixes/corrections.csv (M4, persistent state row)`.

interval covers 0), and is absent for Gemini and Qwen. Audit 3 found that gpt-4.1-mini's gain exists only with the logprob-expected read-out; on the generated integers of the same calls it is negative (§8.1). At FPR 0.01 none of the earlier gains survive: gpt-4.1-mini SR +0.063 [−0.003, +0.170], Gemini top-3 +0.043 [−0.051, +0.156] (both **dropped**; plug-in).

- **Side-task success** (plug-in). 66 of 120 gradual attacks failed their side task. gpt-4.1-mini on successful attacks ($n = 54$): +0.241 [+0.073, +0.336]; failed: +0.067 [−0.067, +0.187]. GPT-4o on successful attacks: +0.051 [−0.053, +0.164]; its pooled gain comes from failed attacks (+0.142 [+0.020, +0.269]).[19]
- **"History kills the gain" holds for one monitor of four.** The DiD is significantly positive only for GPT-4o; Qwen's is not significant; gpt-4.1-mini's is significantly the other way. With all prior diffs, gpt-4.1-mini keeps re-flagging gradual attacks while honest PRs stay low: its max drops to 0.35 and top-3 sum gains +0.315 [+0.156, +0.443] over that weak max.[20]
- **Positional (single-PR) attacks:** max is best for every new monitor (gpt-4.1-mini top-3 − max −0.164 [−0.284, −0.058]).
- Only 5-PR sequences with matching honest runs exist, so longer horizons were tested by streaming over concatenated honest prefixes (§7.2), not by longer attacks.

6.2 MOLE, round 2 (two corpora)

History-free gpt-4.1-mini account-day monitor (MOLE judge prompt and taxonomy) plus MOLE's z-score and peer-fit, 30 workdays, account-level FPR 0.10. Corrected gains (plug-in thresholds):[21]

- GPT-5.3 single-day (30 insiders, 3–4 attack days each), LLM 3rd-max − max: +0.131 [−0.028, +0.287], $p(\leq 0) = 0.058$; top-4 sum +0.086 [−0.021, +0.218]; at FPR 0.05: +0.088 [−0.045, +0.259]. All **not significant** (previously +0.131 [0.11, 0.15]).
- Peer-fit 3rd-max: +0.172 [+0.042, +0.317] (significant on this corpus).
- GPT-5.3 multiday (one attack day): −0.172 [−0.298, +0.041]. DeepSeek-V4 single-day: −0.039 [−0.209, +0.129].
- Accumulators alarm 7–15 workdays later than max. Under MOLE's daily top-K review budget, today's score is best: R@10/day 0.395 for today's gpt-4.1-mini score (MOLE's reported 0.244), vs 0.28 for the running max and 0.31 for the top-3 sum.[22]

6.3 Out-of-sample rule chooser (67 sets)

The 67 sets are the 46 round-1 sets (existing streams and MonitoringBench variants), 12 new persistent-state sets and 9 MOLE sets. The chooser sees the honest calibration half plus $n_{\text{lab}} = 20$ labeled attacks. Round 2 reported +0.023 [0.012, 0.038] for the empirical chooser at FPR 0.01. That number is a winner's curse: the chosen rules over-alarmed (realized FPR 0.0156 vs 0.0142 for always-max), because selecting the rule with the best labeled-attack TPR at its own threshold favours rules whose threshold is low by calibration noise (audit M6).

Once selection noise is separated from threshold noise, no chooser beats always-max (**dropped**). The "always top-3 sum" policy loses (−0.032 nested, family CI [−0.064, −0.004]), and CUSUM with k tuned on the labeled attacks loses −0.051 [−0.092, −0.023] nested.[23]

[19] Source: `results/fixes/corrections.csv (m9)`.
[20] Source: `results/fixes/r2_t1/ps_gains_corrected.csv`.
[21] Source: `results/fixes/corrections.csv (C2, MOLE rows)`.
[22] Source: `results/r2_t1/mole_tables.md`.
[23] Source: `results/fixes/r2_t1/chooser_fixed_summary_task_lab20.csv`.

Chooser (gain over always-max, FPR 0.01)	same calibration (round 2)	nested	FPR-matched
empirical, 20 labels	+0.023 [+0.011, +0.035]	+0.001 [−0.006, +0.009]*	+0.009 [−0.003, +0.023]
empirical, 40 labels	+0.026 [+0.015, +0.038]	−0.001 [−0.013, +0.009]	+0.009 [−0.006, +0.024]
LOSO ridge meta-chooser	+0.019 [+0.007, +0.032]	−0.006 [−0.021, +0.006]	+0.007 [−0.007, +0.025]
mechanism rule	−0.022 [−0.049, +0.005]	−0.017 [−0.047, +0.009]	−0.018 [−0.042, +0.010]
empirical, FPR 0.05	+0.014 [0.005, 0.028]	+0.004 [−0.010, +0.014]	+0.007 [−0.005, +0.016]

Table 8: Rule chooser, 95% CIs from a cluster bootstrap over 14 set families; task-grouped labeled draw. Plug-in thresholds except *, the conformal re-run of the primary nested cell (plug-in +0.001 [−0.014, +0.013]; under the plug-in rule both arms realized 2.04% FPR at nominal 1% because the nested design thresholds on a quarter of the honest set, under the conformal rule 1.01%). Source: `results/fixes/corrections.csv` (M6, M7), `results/fixes/r2_t1/chooser_fixed_summary_task_lab20.csv`, `results/fixes2/cs_overview_conformal_thrfix/`.

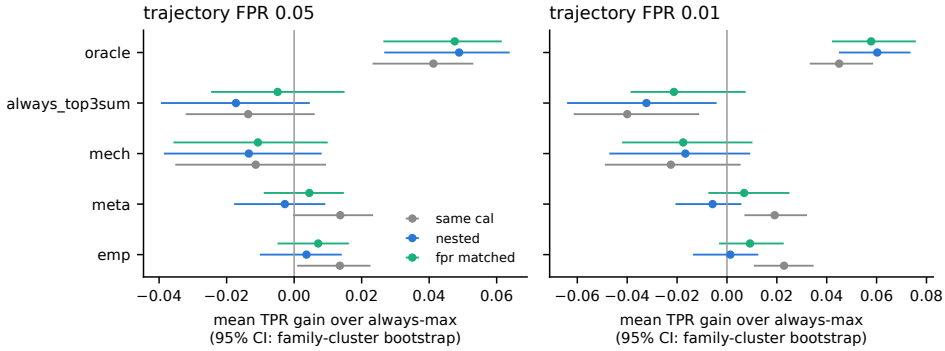


Figure 3: Mean TPR gain over always-max for the chooser variants under same-calibration (grey), nested (blue) and FPR-matched (green) evaluation, 95% family-cluster CIs. Only the oracle (which sees the test attacks) stays above zero.

Length-aware + top-k hybrid (negative, holds). Length-tercile conformal calibration of top-3 sum loses −0.152 [−0.243, −0.040] (family CI) at FPR 0.01; per-set significant losses fall from 36/46 to 19/46 under corrected CIs. The analytic length-aware max remains the only length fix that helps, and only where lengths differ.[24]

6.4 Mechanism correlations are descriptive

Round 1 reported that the top-3 gain tracks the repeat-spike likelihood ratio (Spearman +0.47, 46 sets) and that the CUSUM gain falls with honest spike persistence (−0.67). Those correlations computed the predictor from the same attacks that define the gain and treated correlated sets (full/none, local/cumulative, integer/expected copies) as independent (audit M7). With the predictor estimated on disjoint labeled attacks across 67 sets:[25]

- top-3 sum gain vs repeat-spike LR: +0.30 (naive p 0.013), family-cluster CI [−0.03, +0.45]; family means +0.28 (p 0.33) (**weakened**);
- CUSUM k=0.5 gain vs honest persistence: −0.56, family CI [−0.69, −0.05] at FPR 0.05 (same calibration); under FPR matching [−0.73, +0.02]; family means −0.39 (p 0.16). Directionally consistent, not robust.

[24] Source: `results/fixes/r2_t1/hybrid_summary_corrected.csv`.
[25] Source: `results/fixes/r2_t1/chooser_mechanism_task_lab20.csv`.

The rule built on these properties loses to always-max (Table 8), so "choose the rule from stream properties" is **dropped**.

7 Round 3: accumulation at scale and re-flagging designs

All round-3 gains use the corrected paired bootstrap, nested choosers and reported realized FPR. Pre-specified comparisons: 3rd-max, top-3 sum, CUSUM k=1.5 and the chooser vs max. The round-2 MOLE corpora are "discovery"; the five new corpora are "confirmation".[26]

7.1 MOLE: all eight corpora with one common monitor [C]

The common monitor is a history-free Qwen3-235B-2507 account-day judge (MOLE prompt and taxonomy, compact audit rendering, integer score). Robustness monitors: gpt-4.1-mini with full rendering (GPT-5.3 corpora), MOLE z-score and peer-fit, an activity-only baseline (log events per day) and account-centered variants.

Corpus	n_h/n_a	max TPR	3rd-max − max	top-3 sum − max	CUSUM k=1.5 − max	nested chooser
GPT-5.3 single-day	112/30	0.38	−0.021 [−0.15, +0.09]	−0.044 [−0.15, +0.04]	+0.012 [−0.06, +0.08]	−0.026
DeepSeek-V4 single-day	112/30	0.31	+0.014 [−0.11, +0.13]	+0.077 [−0.08, +0.25]	−0.059 [−0.17, +0.07]	+0.030
Qwen3.6-27B single-day	112/30	0.38	−0.174 [−0.36, −0.02]	−0.085 [−0.25, +0.04]	−0.079 [−0.16, +0.03]	−0.039
Kimi-K2.6 single-day	112/30	0.30	−0.157 [−0.31, +0.03]	−0.096 [−0.22, +0.06]	−0.037 [−0.10, +0.06]	−0.016
GPT-5.3 multiday	112/38	0.35	−0.187 [−0.33, −0.05]	−0.073 [−0.19, +0.07]	−0.004 [−0.08, +0.05]	−0.084
DeepSeek-V4 multiday	114/36	0.18	+0.070 [−0.05, +0.18]	+0.052 [−0.07, +0.20]	+0.065 [−0.05, +0.20]	+0.023
Qwen3.6-27B multiday	113/37	0.23	−0.067 [−0.19, +0.04]	−0.022 [−0.12, +0.07]	−0.081 [−0.14, −0.01]	−0.011
Kimi-K2.6 multiday	114/36	0.17	−0.074 [−0.19, +0.04]	−0.035 [−0.13, +0.08]	−0.018 [−0.08, +0.07]	−0.031
Pooled single-day (RE)			−0.064 [−0.151, +0.024]*	−0.045 [−0.109, +0.019]	−0.030 [−0.073, +0.013]	
Pooled multiday (RE)			−0.060 [−0.160, +0.039]	−0.022 [−0.078, +0.034]	−0.022 [−0.073, +0.030]	

Table 9: Common monitor (Qwen3-235B, integer score), account-level FPR 0.10, paired gains with corrected CIs; realized FPR of max 0.096–0.114. Pooled rows: random-effects meta-analysis over 4 corpora. Plug-in thresholds except *, the conformal re-run (plug-in −0.066 [−0.156, +0.023]); the re-run moved the four single-day 3rd-max cells by ≤ 0.01. Source: `results/r3_t1/mole_tables.md`, `results/fixes2/cs_overview_conformal_thrfix/`.

- **No accumulation rule beats max on any corpus** (no significantly positive gain in Table 9; significant losses on three cells). Mean is worse still (pooled single-day −0.130 [−0.229, −0.031]). The nested chooser stays with max in most splits. Audit 2 reproduced the pooled single-day 3rd-max estimate (−0.073 [−0.166, +0.019]).[27]
- **Provider mix (audit 2, m7).** DeepSeek single-day was scored by a mix of Qwen3-235B providers clustered by account; the logprob provider scores routine days about one point higher. Dropping those accounts: DeepSeek 3rd-max − max −0.012, pooled single-day −0.064 [−0.131, +0.003]. Conclusion unchanged.[28]
- **Horizon sweep** (first 5/10/15/20 workdays, GPT-5.3 single-day, common monitor): no horizon gives a gain above +0.06. Order-statistic rules lose 0.08–0.18 at 5–10 days; at 15–20 days all gains lie between −0.11 and +0.05; CUSUM k=1.5 stays within ±0.01 (point estimates).[29]

[26] Source: `EXPERIMENT_LOG.md (2026-10-22 r3t1 entry)`.
[27] Source: `results/audit2/repro_mole.csv`.
[28] Source: `results/audit2/repro_mole_provider.csv`.
[29] Source: `results/r3_t1/mole_tables.md (horizon sweep)`.

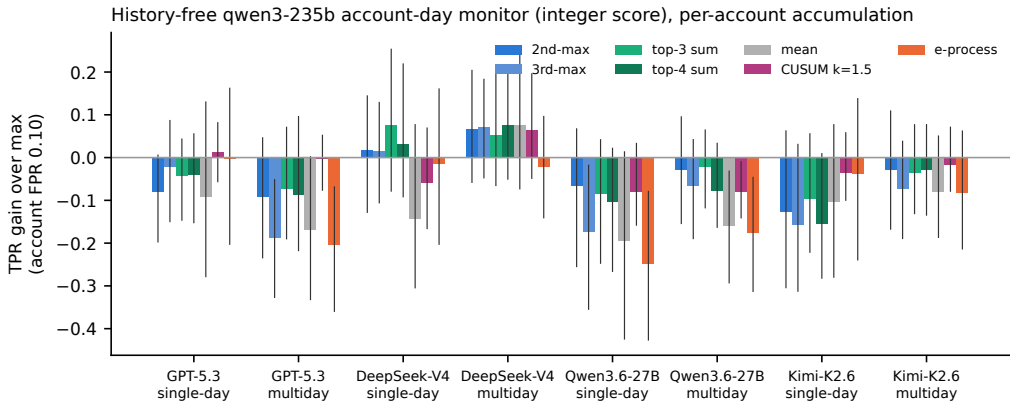


Figure 4: Gain over max at account FPR 0.10 for every accumulator, common monitor, all eight corpora (corrected paired-bootstrap CIs).

- **Daily review budget** (MOLE's R@10/day): today's score is best or within 0.01 of the best on 9 of 11 corpus×monitor cells (today: 0.15–0.43). Exceptions: Kimi single-day (running max 0.18 vs today 0.15) and gpt-4.1-mini on GPT-5.3 multiday (CUSUM k=1.5 0.46 vs 0.43). 3rd-max and top-3 sum are worse everywhere.[30]

The round-2 MOLE gain is not robust [E]. Insider accounts are slightly more active (median events per day 123 on other days vs 116 routine on GPT-5.3 single-day), and the round-2 monitor separates them on non-attack days too (day AUROC 0.65).

Monitor (GPT-5.3 single-day)	max TPR	3rd-max − max	account-centered	without attack days
gpt-4.1-mini (round-2 monitor)	0.31	+0.126 [−0.019, +0.259]	−0.001 [−0.174, +0.126]	+0.048 [−0.058, +0.175] (max 0.125)
Qwen3-235B (common)	0.38	−0.021 [−0.15, +0.09]	−0.027 [−0.15, +0.07]	−0.025
peer-fit	0.17	+0.172 [+0.04, +0.31]	+0.139 [+0.01, +0.33]	+0.123
activity only (no content)	0.00	+0.320 [+0.14, +0.49]	+0.178 [+0.06, +0.31]	+0.232

Table 10: Account-level confound checks at FPR 0.10. gpt-4.1-mini row: audit-2 reproduction (`results/audit2/corrections.csv`, m5); other rows: `results/r3_t1/mole_tables.md`. The activity-only gain is over a degenerate max (TPR 0.00).

The gpt-4.1-mini gain was not significant to begin with and vanishes after account centering; without attack days that monitor barely separates insider accounts (max TPR 0.125). The data cannot separate an account-level confound from noise, so the correct statement is that **the round-2 MOLE gain is not robust**.[31] An activity-only "monitor" with no content reaches 3rd-max TPR 0.32 (the log's "0.37" was a typo; audit 2, m6). Peer-fit keeps a significant gain on GPT-5.3 single-day even after centering, but this does not replicate: pooled over the four single-day corpora peer-fit 3rd-max is +0.022 [−0.097, +0.142].[32]

Mechanism (audit 2, A3) [E]. Across 64 corpus×monitor cells the top-3 gain correlates with repeat-spike informativeness, Spearman +0.297 (naive p 0.017). The cells are 8 corpora × up to 10 dependent monitor variants (centered copies, activity baseline), and the predictor comes from the

[30] Source: `results/r3_t1/mole_tables.md (daily review budget)`.
[31] Source: `results/audit2/corrections.csv (m5)`.
[32] Source: `results/r3_t1/mole_tables.md (meta-analysis)`.

same insider streams. Corpus-cluster 95% CI [−0.04, +0.54]; primary monitors only (29 cells) +0.18 (p 0.36). Descriptive only (**weakened**).[33] For the common monitor on single-day corpora, a repeat high day is barely more likely for insiders than for routine accounts (repeat-spike log-LR −0.02 on GPT-5.3 single-day) while honest persistence is high (0.47).[34]

7.2 Persistent state under continuous monitoring (no reset) [E]

Attack runs are appended after 0–4 random honest test runs; thresholds are set for a matched in-control $\text{ARL}_0 = 5/\alpha$ PRs; detection means an alarm inside the attack run given no prefix alarm. No attack longer than 5 PRs was built: concatenating independent runs would fabricate cross-run continuity that no agent produced.

Set (`current_only`)	ARL_0 target	max (realized ARL_0)	CUSUM k=0.5 (realized ARL_0)	CUSUM k=1.5	2nd-max	top-3 sum
gpt-4.1-mini, gradual	100	0.58 (102)	**0.71** (107)	0.67	0.04	0.00
gpt-4.1-mini, gradual	500	0.48 (no honest alarm)	**0.59** (661)	0.54	0.04	0.00
gpt-4.1-mini, positional	100	**0.65**	0.49	0.65	0.01	0.00
gpt-4.1-mini, positional	500	0.56	0.19	**0.57**	0.01	0.00
GPT-4o (original), gradual	100	0.16 (120)	**0.27** (106)	0.16	0.00	0.00
GPT-4o (original), gradual	500	0.01 (no honest alarm)	**0.19** (2,387)	0.01	0.00	0.00

Table 11: Detection rate under continuous monitoring. Sources: `results/r3_t1/ps_longhorizon.md`, `results/r3_t1/ps_longhorizon_arl.csv`. No CIs were computed in round 3; round 4 adds them (§8.3). "No honest alarm": the integer-score max threshold never fired on the honest test stream, so its realized ARL_0 is unresolved.

- For history-free gpt-4.1-mini on gradual attacks, CUSUM k=0.5 has the higher point estimate: 0.71 vs 0.58 at $\text{ARL}_0 = 100$ PRs (realized ARL_0 107 vs 102) and 0.59 vs 0.48 at 500. GPT-4o: 0.27 vs 0.16 at 100. Round 4 re-ran this with CIs: +0.132 [−0.086, +0.296], not significant (§8.3).
- On positional (single-PR) attacks, CUSUM k=0.5 loses (0.49 vs 0.65); k=1.5 matches max.
- Non-forgetting accumulators (k-th largest, top-k sum over the whole stream) collapse to ≈ 0, so a deployed accumulator has to forget (CUSUM) or be windowed. Round 4 finds that forgetting rules do not beat max either (§8.3).
- Caveats: point estimates without CIs (round 3); at $\text{ARL}_0 = 500$ the max threshold is not matched; Gemini and Qwen streams give low or mixed detection (Gemini history-free ≤ 0.14 for every rule).

7.3 Re-flagging as a monitor design target [E]

Round 2 suggested that what helps accumulation is attack re-flagging (a high repeat-spike LR), not history per se. Round 3 tested monitor designs that target it, all with gpt-4.1-mini, as matched pairs differing only in the variable under test: *history-free*; *history-aware* (plain); *reflag* = history-aware + an explicit rule to re-raise suspicion when the current unit continues earlier suspicious work; *ledger* = no history but the monitor's own ledger of open suspicions, with a *ledger-null* control (identical prompt, notes always empty); two samples at $T = 1$ (each alone, their mean, or both as separate units).

[33] Source: `results/audit2/corrections.csv (A3)`.

[34] Source: `results/r3_t1/mole_tables.md (per-corpus stream features)`.

Design	run-max AUROC	repeat-spike log-LR	max @0.05	top-3 @0.05 (gain over max)	max @0.01	top-3 @0.01 (gain)
history-free	0.897	+1.90	0.58	**0.74** (+0.16 [+0.02, +0.23])	0.50	0.53 (+0.03 [−0.03, +0.22])
history-aware	0.831	+1.91	0.37	0.67 (+0.30 [+0.16, +0.42])	0.21	**0.57** (+0.35 [+0.24, +0.44])
reflag	0.806	+2.00	0.40	0.56 (+0.16 [+0.05, +0.32])	0.14	0.44 (+0.30 [+0.12, +0.39])
ledger	0.856	+1.45	0.55	0.64 (+0.08 [−0.03, +0.19])	0.23	0.18 (−0.05 [−0.11, +0.19])
ledger-null	0.860	+3.63	0.37	0.58 (+0.21 [+0.03, +0.33])	0.08	0.27 (+0.19 [+0.08, +0.33])
sample0 (T=1)	0.816	+2.90	0.45	0.50 (+0.05 [−0.01, +0.22])	0.26	0.26 (+0.00 [−0.05, +0.14])
sample1 (T=1)	0.858	+2.42	0.49	0.63 (+0.14 [+0.04, +0.21])	0.43	0.52 (+0.09 [+0.03, +0.19])
sample mean	0.868	+1.92	0.57	0.72 (+0.16 [+0.03, +0.24])	0.45	0.45 (+0.00 [−0.05, +0.21])
sample units	0.838	+1.33	0.53	0.66 (+0.13 [+0.05, +0.24])	0.34	0.51 (+0.17 [+0.07, +0.23])
length only (− log chars)	0.585	+0.32	0.13	0.28 (+0.15 [+0.10, +0.33])	0.05	0.16 (+0.11 [+0.06, +0.16])

Table 12: Persistent state, gradual attacks (155 honest / 120 attack), run-level FPR 0.05 and 0.01; top-3 sum on whitened scores; corrected paired-bootstrap CIs; plug-in thresholds. Source: `results/r3_t1/reflag_tables.md` (ps:gradual).

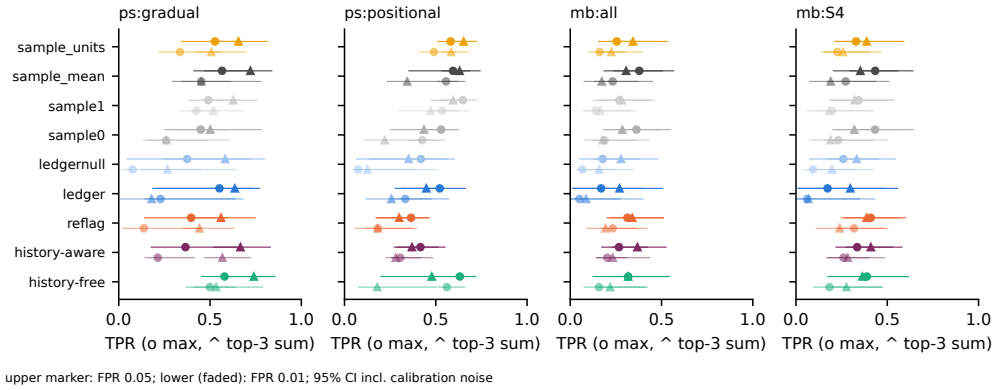


Figure 5: TPR of max (o) and whitened top-3 sum (∧) per design; upper marker FPR 0.05, lower 0.01; 95% CIs include calibration noise.

- **Re-flagging is not a better design.** The explicit reflag rule lowers run-max AUROC relative to the plain history-aware monitor (0.806 vs 0.831) and leaves the repeat-spike LR unchanged (2.00 vs 1.91). The best rule at 5% FPR is history-free whitened top-3 sum (0.74). Audit 2 reproduced it exactly (0.742 vs 0.581, +0.161 [+0.026, +0.233]) but found that it depends on the whitening: with raw logprob-expected scores the top-3 sum gains only +0.031 [+0.001, +0.070], and at FPR 0.01 +0.037 [−0.019, +0.213].[35]
- At 1% FPR the best rules are close with overlapping CIs: history-aware top-3 0.57, history-free top-3 0.53, history-free max 0.50.
- **Ledger.** The carried notes raise max-type detection relative to ledger-null (0.55 vs 0.37), but the ledger monitor's honest scores grow with diff length (Spearman +0.71 within honest PRs), and it is not better than history-free.[36]
- **Two samples.** Each T=1 sample has a weaker max than T=0 (0.45 / 0.49 vs 0.58); their mean recovers it (top-3 0.72). Using both samples as units gives a small significant accumulation gain on *positional* attacks at 1% FPR: top-3 +0.09 [+0.03, +0.13] over that stream's max.
- **Nested chooser** gains +0.05 to +0.31 over max where top-k clearly wins (persistent-state gradual) and stays within ±0.07 of max elsewhere, at realized FPR close to nominal.
- **MonitoringBench** (150 honest / 150 attacks): the best TPR at 5% FPR is 0.39 on all attacks and 0.46 on S4 (history-aware 2nd-max in both); history-free max is 0.32 / 0.39. The reflag

[35] Source: `results/audit2/issues.csv (m8)`.
[36] Source: `results/r3_t1/ps_length_check.csv`.

rule lowers the repeat-spike LR vs the plain history-aware prompt (0.45 vs 0.58). Gains over max reach +0.14 to +0.15 only where that design's own max is weak (ledger-null all attacks, 2nd-max +0.14 [+0.06, +0.23]).[37]

Validity checks: length. Gradual-attack diffs are about 4× shorter than honest diffs (median 13,427 vs 53,602 characters), and a length-only "monitor" with 2nd-max reaches TPR 0.40 at 5% FPR. Monitor scores *rise* with diff length within honest PRs (gpt-4.1-mini +0.09, Qwen +0.51, ledger +0.71), so length works against the monitors; it also explains Qwen's high honest persistence in round 2. Length-matched subsets (25/25) are too small to conclude anything (realized FPR ≈ 0.15). On MonitoringBench, attacks are longer (stream-length AUROC 0.72); the opus-4-5-only and length-matched subsets give the same qualitative picture.[38]

8 Round 4: nine monitor families, a stronger MOLE monitor, unreset streams with CIs

Track `r4cs` (October 22; OpenRouter $143.53). Round 4 asked whether the cross-session gains of rounds 1–3 are a property of history-free per-PR monitoring or of a few monitors, whether a stronger account-day monitor creates an accumulation gain on MOLE, and whether the round-3 unreset-stream result survives CIs. Each analysis script states its pre-specification in its docstring (§4.3). Audit 3 re-ran every analysis and re-implemented the headline contrasts independently; all headline numbers reproduced, and four interpretations were corrected (below).[39]

8.1 Persistent state: nine history-free monitors with identical inputs

Design. 153 honest / 117 gradual runs that have five scored PRs under every monitor (5 apps). Six new monitors (gpt-4.1, Claude Haiku 4.5, Gemini 2.5 Pro, gpt-5-mini, Claude Haiku 5.5, GPT-6 Luna) were added to the round-2 three, all with the round-2 `current_only` PR-review prompt and identical inputs. Only API-side settings differ (Gemini 2.5 Pro's 1,024-token reasoning budget, low reasoning effort for gpt-5-mini and Luna, larger completion caps where needed). One difference was not disclosed in the round-4 log: gpt-4.1-mini, Qwen3-235B and gpt-4.1 are read out as the logprob-expected score (continuous), the other six as the generated integer (audit 3).[40] 40 iid 50/50 honest splits; gains use the joint bootstrap clustered by app (B = 2,000). GPT-4o's original round-1 scores are a reference and are not pooled.

- **Pre-specified primary [C]: no pooled gain.** Top-3 sum − max pooled over the nine identical-input monitors is −0.066 [−0.147, +0.015] (plug-in −0.070 [−0.144, +0.015]); HKSJ [−0.209, +0.077]; $I^2 = 0.88$; 95% prediction interval for a new monitor [−0.55, +0.42]. The audit's independent implementation gives −0.070 [−0.152, +0.012] and a leave-one-app-out jackknife −0.069 [−0.206, +0.068] (plug-in).[41] The interval is conditional on these nine monitors; the prediction interval is the relevant range for another monitor.
- **Per monitor [E].** With Holm over the nine monitors (app bootstrap), only GPT-6 Luna's loss is significant (−0.422, $p_{\text{Holm}} = 0.005$); Claude Haiku 5.5's −0.247 has $p_{\text{Holm}} = 0.07$. No monitor

[37] Source: `results/r3_t1/reflag_tables.md (mb sections)`.
[38] Sources: `results/r3_t1/ps_length_check.csv`; `results/r3_t1/reflag_tables.md`.
[39] Source: `results/audit3/corrections.csv`.
[40] Source: `results/audit3/cs_ps_granularity.csv`.
[41] Sources: `results/fixes2/headline_threshold_rerun.csv`; `results/audit3/cs_ps_jackknife.csv`.

Monitor	read-out	run-max AUROC	max TPR (realized FPR)	top-3 sum − max [app CI]	3rd-max − max [app CI]
gpt-4.1-mini (r2)	logprob	0.894	0.561 (0.046)	+0.171 [+0.008, +0.274]	+0.026 [−0.233, +0.274]
Gemini 2.5 Flash (r2)	integer	0.623	0.211 (0.046)	+0.036 [−0.073, +0.157]	−0.063 [−0.283, +0.130]
Qwen3-235B (r2)	logprob	0.737	0.446 (0.050)	−0.059 [−0.155, +0.088]	−0.142 [−0.328, +0.130]
gpt-4.1	logprob	0.818	0.515 (0.044)	−0.138 [−0.299, +0.166]	−0.320 [−0.475, −0.112]
Claude Haiku 4.5	integer	0.707	0.230 (0.052)	+0.106 [−0.094, +0.253]	−0.002 [−0.236, +0.167]
Gemini 2.5 Pro	integer	0.674	0.131 (0.041)	+0.020 [−0.152, +0.145]	−0.022 [−0.297, +0.046]
gpt-5-mini	integer	0.797	0.460 (0.052)	−0.061 [−0.158, +0.068]	−0.346 [−0.574, −0.131]
Claude Haiku 5.5	integer	0.894	0.727 (0.051)	−0.247 [−0.420, −0.076]	−0.650 [−0.776, −0.558]
GPT-6 Luna	integer	0.912	0.748 (0.049)	−0.422 [−0.506, −0.281]	−0.659 [−0.780, −0.531]
Pooled, 9 monitors (RE, joint app bootstrap)				**−0.066** [−0.147, +0.015]	−0.250 [−0.380, −0.139]
GPT-4o (original, reference)	integer	0.634	0.148 (0.044)	+0.086 [−0.004, +0.218]	+0.040 [−0.062, +0.161]

Table 13: Round 4, gradual attacks, run-level FPR 0.05, conformal thresholds. Intervals are per-monitor and unadjusted. Source: `results/fixes2/cs_ps_multi_conformal_thrfix/ps_tables.md` (plug-in version: `results/r4_cs/ps_tables.md`); read-out from `results/audit3/cs_ps_granularity.csv`.

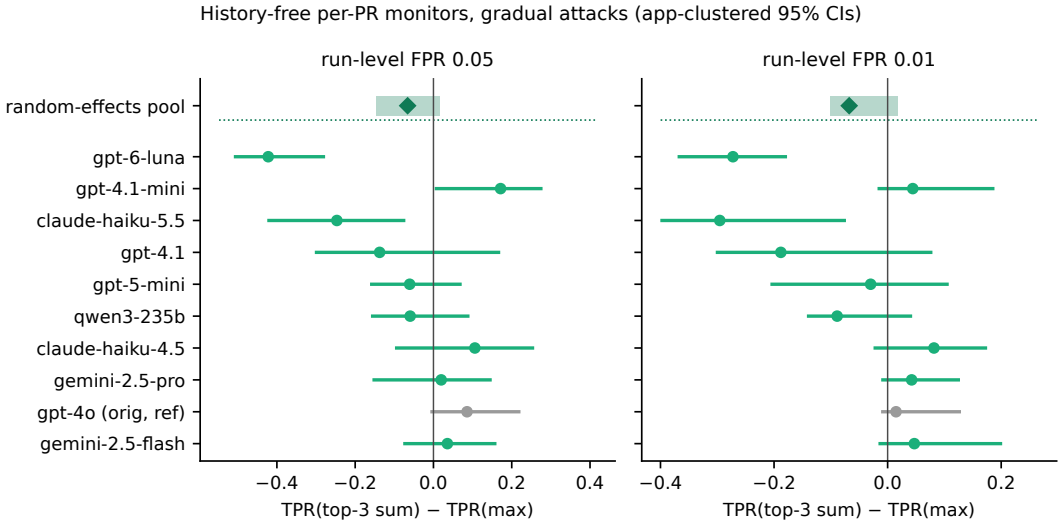


Figure 6: Top-3 sum − max per monitor and the random-effects pool, run-level FPR 0.05 (left) and 0.01 (right), app-clustered 95% CIs, conformal thresholds (generated by `tracks/r4_cs/ps_multi.py` under the conformal rule). The FPR-0.01 panel is descriptive: with about 76 calibration runs per split every threshold sits at the largest calibration statistic.

has a gain that survives Holm. Unadjusted, gpt-4.1-mini's app interval excludes 0 under the conformal rule ([+0.008, +0.274]; plug-in [−0.006, +0.275]).

- **gpt-4.1-mini's gain is a read-out effect [E].** On the generated integers of the same gpt-4.1-mini calls, top-3 sum − max is −0.034 [−0.052, −0.009], against +0.141 [+0.007, +0.249] for the logprob read-out in the same (audit, plug-in) implementation. gpt-4.1-mini is also not a weak monitor (run-max AUROC 0.894, tied second). The round-4 log's reading that the gain belongs to "weak-to-mid monitors" therefore fails. The pooled estimate barely depends on read-out (−0.074 → −0.083 with integer read-outs for all three logprob monitors; plug-in).[42] The remaining positive point estimates are GPT-4o (integer, +0.086 [−0.004, +0.218] on this run set) and Claude Haiku 4.5 (+0.106, n.s.).
- **Other contrasts.** 3rd-max − max pooled is −0.250 [−0.380, −0.139] (secondary [C]); after Holm three monitors remain significantly negative (gpt-5-mini, Haiku 5.5, Luna; plug-in). At FPR 0.01 the pooled top-3 contrast is −0.068 [−0.101, +0.017] (plug-in −0.099); it is descriptive

[42] Source: `results/audit3/cs_ps_granularity.csv`.

only, since every threshold sits at the largest of about 76 calibration statistics [E]. Plug-in robustness checks [E]: leave-apps-out calibration −0.076 [−0.134, +0.012] (max's realized FPR 0.049–0.109 across monitors), length-residualized scores −0.031 [−0.136, +0.053].

- **Why the strongest monitors lose [E] (post hoc).** They put a near-ceiling score on the single payload PR, so max is already high (0.73–0.75), while their honest scores rise with diff length (Spearman +0.47 to +0.67 for gpt-4.1, gpt-5-mini, Haiku 5.5 and Luna), which inflates the honest top-3 tail. Across the nine monitors the gain falls with that correlation (Spearman −0.73, naive $p = 0.025$, $k = 9$).
- **GPT-4o on the common run set** is +0.086 [−0.004, +0.218], below the round-1 +0.134 [+0.024, +0.217] (different run filter, 40 vs 10 splits, app vs iid clusters).

Mechanism (pre-specified, [C]). For each monitor, y = top-3 − max on attacks from one half of the apps and x = repeat-spike log-LR from attacks of the other half, averaged over 20 app partitions; the joint bootstrap reuses the same run weights for x, y and every monitor. The cross-fitted Spearman correlation across the nine monitors is +0.065 [−0.341, +0.460] (app clusters; not cross-fitted +0.100 [−0.417, +0.567]). Across all 42 corpus×monitor cells (persistent state and MOLE) it is +0.016, two-way cluster CI [−0.54, +0.59]. Claude Haiku 5.5 has the highest repeat-spike LR (2.72) and the second-largest loss. With $k = 9$ the test is inconclusive: it removes the round-3 support for the mechanism but is not evidence of no relationship.[43]

8.2 MOLE with a stronger day monitor

A search of Hugging Face and GitHub found no public release with more than five sessions per run and matched honest runs, so round 4 re-used MOLE's four single-day corpora (112 routine vs 30 solo insider accounts, 3–4 attack days each, 30 workdays). The renders were rebuilt with the round-2 code (account-day tables identical to rounds 2–3) and scored by GPT-6 Luna on the round-3 compact input, the same input as the common Qwen3-235B monitor. A gpt-4.1 run covers a budget subset (GPT-5.3 single-day: all 30 insiders and 50 random routine accounts). MOLE results use plug-in thresholds (not re-run; $\alpha n = 5.6$ at account FPR 0.10, so the plug-in excess is small: the pooled Qwen row moved by 0.002 under the conformal rule).

- **Day level [E].** Luna separates attack days from routine days better than Qwen (AUROC 0.69–0.80 vs 0.57–0.69) and also scores insiders' non-attack days somewhat higher than routine days (AUROC 0.60–0.64 vs 0.53–0.58).[44]
- **Primary [C]: no accumulation gain.** 3rd-max − max at account-level FPR 0.10, pooled over the four corpora, is +0.047. The four corpora share all 142 accounts and the same 30 insiders, so they are not independent; a joint account bootstrap (same account weights in every corpus) gives [−0.026, +0.125]. The round-4 log's interval [−0.019, +0.114] treated the corpora as independent. Per corpus: −0.019 to +0.087, none significant. Qwen on the same accounts: −0.061 [−0.150, +0.028]. Top-3 sum − max: +0.030 [−0.008, +0.091] (joint).[45]
- **Luna's detection comes from attack days [E].** With attack days removed, Luna's max TPR at account FPR 0.10 falls to chance in every corpus (0.08–0.18 at realized FPR 0.10–0.11; GPT-5.3 single-day 0.328 → 0.120). Account centering gives TPR 0.000, which is below chance: on a bounded 0–10 scale, a monitor that saturates (insider account medians ≈ 7.5) cannot be centered meaningfully. The round-4 log's reading that Luna's insider signal is "an account-level

[43] Source: `results/r4_cs/mechanism.md`.
[44] Source: `results/r4_cs/mole_tables.md`.
[45] Source: `results/audit3/cs_mole_joint.csv`.

offset" therefore fails. The mild offset visible at day level does not drive account-level detection. Without attack days, 3rd-max − max is +0.041 [−0.009, +0.102] (joint), a comparison of two near-chance rules.[46]

- **Activity-only control** [E]**.** On GPT-5.3 single-day an activity-only score again gives a large 3rd-max − max (+0.338 [+0.164, +0.514]) over a degenerate max (TPR 0), and no effect on the other three corpora.
- **gpt-4.1 subset** [E] (underpowered: 25 calibration accounts per split; realized FPR of max 0.138 at nominal 0.10). Max 0.371, 3rd-max − max +0.093 [−0.075, +0.253]. The realized FPR differs by rule (3rd-max 0.115) in max's favour, so the contrast is if anything understated.
- **Daily review budget** [E] (R@10/day, no CIs): today's score is the best rule in 9 of 11 monitor×corpus rows (e.g. Luna on GPT-5.3 single-day 0.432 vs 0.296 for the running max). Exceptions: DeepSeek with Luna (CUSUM k=1.5 0.243 vs 0.234) and Kimi with Qwen (running max 0.184 vs 0.153).

8.3 Unreset streams with confidence intervals

Pre-specified in `ps_arl.py`: no reset across runs; each attack run follows U{0..4} honest runs; thresholds matched to ARL_0 on calibration runs (ties randomized by a 10^{-6} jitter); point estimate = mean over 10 splits; CI = 200 run-bootstrap replicates of the whole procedure with recalibrated thresholds, plus an app-cluster bootstrap at $\mathrm{ARL}_0 = 100$. These thresholds are set by ARL matching, so the threshold convention does not apply.

- **P3a** [C] **(round-3 claim with CIs).** gpt-4.1-mini, $\mathrm{ARL}_0 = 100$ PRs: CUSUM k=0.5 0.718 vs max 0.586, +0.132 [−0.086, +0.296] (app clusters [−0.09, +0.32]). The point estimate reproduces round 3; the difference is not significant. The interval is conservative (each replicate uses one split while the point averages 10), but a split-averaged approximation, [−0.019, +0.284], is not significant either. $\mathrm{ARL}_0 = 500$: +0.120 [−0.076, +0.248].[47]
- **P3b** [C]**.** Windowed top-3 ($W = 5$) − max, gpt-4.1-mini: +0.123 [−0.084, +0.266].
- **Pooled over the nine monitors, $\mathrm{ARL}_0 = 100$** [C]**.** Windowed top-3 −0.116 [−0.178, −0.053] (survives Holm over the two pre-specified pooled contrasts); CUSUM k=0.5 −0.032 [−0.109, +0.049]. Exploratory [E]: EWMA ($\lambda = 0.2$) −0.088 [−0.162, −0.009], Shiryaev–Roberts −0.061 [−0.133, +0.011], CUSUM k=1.5 +0.004 [−0.019, +0.020].
- **Nested best forgetting rule** [E] (chosen on attacks from one half of the apps, evaluated on the other): +0.015 [−0.010, +0.062] at $\mathrm{ARL}_0 = 100$. Max is not shown to be worse, but gains up to about +0.06 cannot be excluded. At $\mathrm{ARL}_0 = 500$ it is +0.039 [+0.007, +0.070], but that cell is unmatched: max raises no honest alarm in 47% of splits while CUSUM-type rules over-alarm out of sample (median realized ARL_0 168–351 PRs), which favours the forgetting rules.
- The strong integer-score monitors (Luna, Haiku 5.5, gpt-5-mini) do best with CUSUM $k = 1$–2, which behaves like max; low-k CUSUM, EWMA and windowed sums lose 0.1–0.4 for them.

8.4 Reasoning leakage (audit 3)

Agent reasoning is nearly label-revealing in these datasets (e.g. on MonitoringBench, the amount of visible reasoning separates honest from attack runs almost perfectly). Audit 3 checked every

[46] Sources: `results/audit3/corrections.csv`; `results/r4_cs/mole_matched.csv`.
[47] Sources: `results/r4_cs/arl_tables.md`; `results/audit3/cs_arl_ci_check.csv`.

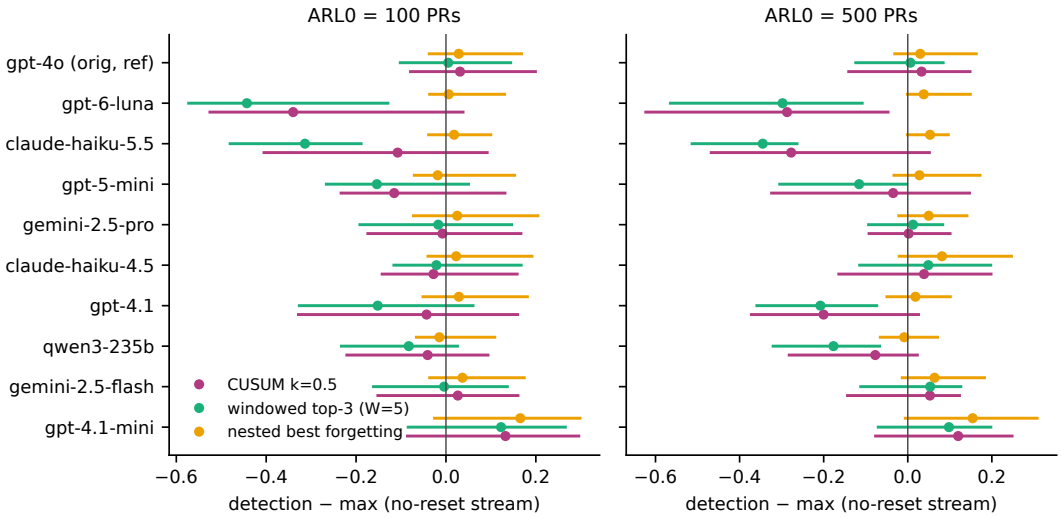


Figure 7: Unreset streams: detection minus max at matched $\mathrm{ARL}_0 = 100$ (left) and 500 (right) PRs, per monitor, for CUSUM k=0.5, windowed top-3 ($W = 5$) and the nested best forgetting rule, with run-bootstrap 95% CIs. The $\mathrm{ARL}_0 = 500$ panel is not matched (see text).

monitor scored in this repository: none receives agent reasoning (it is rendered only with an option no track sets), and no text-only assistant turn appears among the visible steps of the MonitoringBench trajectories. The history-aware vs history-free comparisons in this report are therefore not reasoning leakage. Visible step count remains label-correlated on MonitoringBench (AUROC 0.70), which history-aware and trajectory-level rules can use. Externally computed scores (the persistent-state GPT-4o natives) cannot be checked.[48]

9 What holds, what was weakened, what was dropped

Audit 1 found 2 critical, 9 major and 13 minor issues across all tracks; those touching Idea 1 are C2, M4, M6, M7, M8, M9, m1, m7, m9, m10, m12.[49] Audit 2 re-implemented the round-3 analyses independently (randomized plug-in thresholds, task-grouped splits, a joint cluster bootstrap with thresholds recomputed); every round-3 Idea-1 headline reproduced within split and bootstrap noise, and the issues affecting this report are A3, m5, m6, m7 and m8.[50] Audit 3 re-ran the round-4 track from committed code (identical tables) and re-implemented its headline contrasts; it corrected four interpretations (read-out, Holm count, MOLE account sharing, MOLE account-level reading) and several minor statements.[51] The threshold-convention re-run changed no Idea-1 verdict. Figure 8 shows the size of the audit-1 inference correction.

Holds

- Max-type rules (max, 2nd-max, length-aware max) match or beat every accumulator on most real per-step stream sets (86% / 67% / 83% of aksh-n / LinuxArena / MRT sets at FPR 0.01);

[48] Source: `results/audit3/leak_reasoning.csv`.
[49] Source: `results/audit/issues.csv`.
[50] Source: `results/audit2/issues.csv`.
[51] Source: `results/audit3/corrections.csv`.

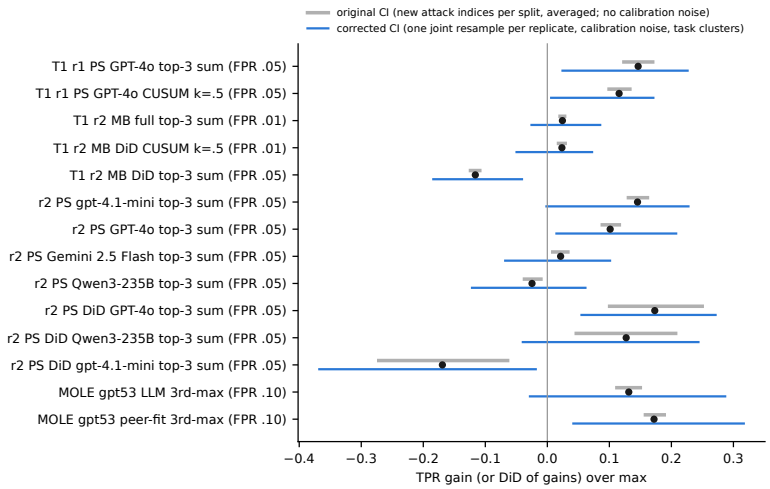


Figure 8: Audit-1 correction of the gain CIs: headline gains over max with the original CIs (grey; new attack indices per split, averaged; no calibration noise) and the corrected CIs (blue; one joint resample per replicate, calibration noise, task clusters). Both use the plug-in threshold rule; the figure documents the bootstrap correction, not current estimates (e.g. GPT-4o top-3 is now +0.134 [+0.024, +0.217] under the conformal rule).

textbook CUSUM k=0.5, SR, long-window GLR and e-processes lose almost everywhere.[52]

- Cross-session accumulation of history-free GPT-4o per-PR scores: +0.134 [+0.024, +0.217] (round 1, conformal), +0.101 [+0.015, +0.208] (paired filter, plug-in); history-aware GPT-4o loses the gain (DiD +0.173 [+0.055, +0.271]). It is one monitor: across nine monitor families the pooled gain is −0.066 [−0.147, +0.015] (round 4).
- Non-forgetting top-k collapses under continuous no-reset monitoring. The round-3 CUSUM point estimate for gpt-4.1-mini (0.71 vs 0.58 at $\mathrm{ARL}_0 = 100$) reproduces in round 4, but with CIs it is +0.132 [−0.086, +0.296], not significant.
- MOLE: no accumulation rule beats max for the common monitor on any of 8 corpora (pooled single-day 3rd-max −0.064 [−0.151, +0.024]); today's score is the best (or within 0.01 of the best) daily-review rule on 9 of 11 cells (round 3) and the best rule on 9 of 11 rows (round 4).
- MOLE with a stronger day monitor (GPT-6 Luna): no gain, +0.047 [−0.026, +0.125] (joint account bootstrap).
- Round 4 pooled windowed top-3 sum loses to max on unreset streams (−0.116 [−0.178, −0.053]); only Luna's per-monitor top-3 loss survives Holm.
- CUSUM's reference value is critical ($k \approx 1.5$–$2 \approx$ max on whitened LLM scores; small k over-alarms); e-process Ville thresholds over-alarm on dependent streams.
- Length-aware max on MRT (+0.365 [+0.127, +0.445]), conditional on unadapted attack lengths.
- Weak history-free monitor (gpt-4.1-nano): top-3 sum +0.026 [+0.001, +0.058] at FPR 0.01.
- Length-tercile hybrid is a loss (−0.152 [−0.243, −0.040]).
- Positional (single-PR) attacks: max is best (gpt-4.1-mini top-3 −0.164 [−0.284, −0.058]).

[52] Source: `results/t1/tables/r1_tpr.md`.

Weakened

- History-free gpt-4.1-mini PR gain: +0.145 [+0.007, +0.230] (conformal; plug-in [−0.001, +0.228]); the app-cluster interval covers 0; it exists only with the logprob-expected read-out (integer read-out of the same calls −0.034 [−0.052, −0.009]); significant on side-task-successful attacks (+0.241 [+0.073, +0.336], plug-in); the round-3 whitened top-3 gain (+0.16 [+0.02, +0.23]) shrinks to +0.031 with raw scores.
- Repeat-spike mechanism: inconclusive across monitors (cross-fitted Spearman +0.065 [−0.341, +0.460], $k = 9$).
- "Max is as good as any forgetting rule" on unreset streams (round-4 log): non-significance only; nested best forgetting rule +0.015 [−0.010, +0.062] at $\mathrm{ARL}_0 = 100$.
- GPT-4o CUSUM gain +0.116 [+0.006, +0.171]; app-cluster interval crosses zero. GPT-4o's pooled gain comes from side-task-failed attacks.
- "History removes the gain": significant for 1 of 4 monitors (GPT-4o), reversed for gpt-4.1-mini.
- Mechanism correlations (repeat-spike LR, honest persistence): descriptive; family/corpus-cluster CIs include or approach zero (67 sets: +0.30 [−0.03, +0.45]; MOLE: +0.297 [−0.04, +0.54]).
- Re-flagging under history on MonitoringBench: +0.03 [−0.03, +0.09] with matched prompts.
- Absolute MonitoringBench TPR: about 12% like-for-like (opus-4-5) at 1% FPR; realized FPR averages 1.28× nominal at $\alpha = 0.01$.
- "Round-2 MOLE gain was an account-level confound": restated as not robust (never significant; vanishes after centering).
- Max alarms at the ideal flag step: partly by construction.

Dropped or reversed

- MonitoringBench full-mode top-3 sum gain at FPR 0.01 (+0.024 [−0.025, +0.085]) and the CUSUM DiD (+0.024 [−0.050, +0.072]).
- "Accumulators gain more with history" (negative DiD at FPR 0.05): instruction artifact; near zero with matched prompts.
- "History-free max beats history-aware max" on MonitoringBench: reversed with matched prompts (−0.091 [−0.186, +0.038]).
- Persistent-state gains for Gemini 2.5 Flash and Qwen3-235B, and all FPR-0.01 persistent-state gains from round 2.
- Qwen's history DiD (+0.127 [−0.040, +0.244]).
- MOLE round-2 3rd-max gain (+0.131 [−0.028, +0.287]) as a positive result.
- Rule chooser gain (+0.023 → +0.001 nested) and the practitioner lesson "estimate stream properties, then pick the rule".
- "Deployed (no-reset) monitoring needs a forgetting accumulator" as a demonstrated advantage over max (round 3 summary); only the collapse of non-forgetting rules stands.
- "The cross-session gain is a property of weak-to-mid monitors" (round-4 log): gpt-4.1-mini's gain is a read-out effect and gpt-4.1-mini is not weak.
- "Luna's MOLE insider signal is mostly account-level" (round-4 log): without attack days its detection is at chance; centering fails at the score ceiling.

Sources for this section: `results/fixes/corrections.csv`, `results/audit2/corrections.csv`, `results/audit3/corrections.csv`, `results/fixes2/headline_threshold_rerun.csv` and the tables cited above.

10 Conclusions

1. **Keep max as the default decision rule, with a length-aware null where honest and attack lengths differ.** At a matched budget, textbook sequential tests with default parameters lose to max-type rules on almost every per-step stream set, and no data-driven rule selection we tried beats always-max out of sample.
2. **Cross-session accumulation is not a general gain.** Across nine history-free per-PR monitors with identical inputs the pooled top-3 − max is -0.066 $[-0.147, +0.015]$, with strong heterogeneity: GPT-4o gains, gpt-4.1-mini gains only with its logprob read-out, and the strongest monitors lose up to 0.42. Whether accumulation helps is a property of the particular monitor and score read-out, so it must be measured per monitor at a matched budget before it is deployed.
3. **In a deployment without resets, do not use non-forgetting order statistics.** Max is not shown to be worse than CUSUM, EWMA or Shiryaev–Roberts at matched ARL_0, and windowed sums lose. A forgetting rule with a large reference ($k \approx 1$–2) behaves like max and is a safe choice.
4. **Insider-style account monitoring does not benefit** from per-account accumulation of a content monitor across eight MOLE corpora, also with a stronger monitor; review today's highest-scoring account-days.
5. **The history double-counting hypothesis is not supported** on MonitoringBench, and the earlier opposite effect was an instruction artifact. Prompt wording moves detection as much as history does, so history-mode comparisons must hold the instructions fixed. Score read-out (logprob-expected vs generated integer) can also flip a conclusion, so it must be held fixed and reported.
6. **If using CUSUM on LLM scores, tune k to the score distribution** ($k \approx 1.5$–2 on whitened scores), calibrate thresholds with a rule whose out-of-sample FPR is exact (the conformal randomized rule), and report realized FPR.

11 Limitations

- **Small honest sets** (112–550 per set): FPR-0.01 thresholds are noisy (realized FPR up to 2.3× nominal under the plug-in rule) and many contrasts are inconclusive rather than negative. Several round 1–3 tables were not re-run under the conformal rule and keep plug-in thresholds.
- **Short cross-session sequences** (5 PRs, 5 apps): long-horizon results splice honest prefixes before 5-PR attacks; app-clustered intervals rest on 5 clusters and are coarse, so iid intervals are given alongside. No public multi-session release with more than five sessions and matched honest runs was found.
- **Read-out heterogeneity.** Three of the nine round-4 monitors were read out as logprob-expected scores and six as integers; the pooled estimate is insensitive to this, the gpt-4.1-mini gain is not.
- **MOLE dependence.** The four single-day corpora share all accounts; only the joint account bootstrap is valid for pooled MOLE results. The gpt-4.1 MOLE subset is underpowered and runs at realized FPR 0.14.
- **Label and surface confounds.** MonitoringBench agent-model mismatch (honest all opus-4-5); persistent-state diff length (attacks 4× shorter); MOLE insider activity; side-task failures counted as positives in pooled persistent-state numbers.
- **Heuristic onset labels** (MonitoringBench execution step, aksh-n ideal-flag step).

- **Monitor coverage.** Rounds 1–3 used cheap monitors and the round-3 design comparison uses gpt-4.1-mini alone; round 4 adds six families on persistent state and GPT-6 Luna on MOLE.
- **Static attacks.** No attacker adapted to the detector (Idea 3 covers adaptive attackers).
- **Analyst degrees of freedom.** Round-2 chooser variants were chosen after seeing results; round 3 pre-specified its comparisons and separated discovery and confirmation corpora; round 4 pre-specified its primaries in script docstrings, two of which were committed after the pre-commit (written before their outputs, per session transcripts).

12 Next steps

Build persistent-state campaigns longer than 5 PRs with matching honest sequences, to test forgetting accumulators with more power; analyse `idacy/ca-pse-iclr2027` (13,311 runs, 8 monitors) once its history mode is documented; re-run the round 1–3 tables that still use plug-in thresholds; combine the length-aware null with Idea 2's partially pooled per-context baselines.

Reproduction

Code: `tracks/t1_matched_budget/`, `tracks/r2_t1/`, `tracks/r3_t1/`, `tracks/r4_cs/`, `tracks/fixes/`, `tracks/audit2/`, `tracks/audit3/`, `tracks/fixes2/`, and the library `amd/` (detectors, evaluation incl. `amd.evaluation.paired` and the conformal threshold rule, scoring). Outputs: `results/t1/`, `results/r2_t1/`, `results/r3_t1/`, `results/r4_cs/`, `results/fixes/`, `results/audit/`, `results/audit2/`, `results/audit3/`, `results/fixes2/`. Narrative log: `EXPERIMENT_LOG.md`. Repository: https://github.com/gregkocher/agent-monitor-detection (private).

13 Errata relative to the October 22 version

Table 14 lists the statements of the October 22 version that audit 3 or the threshold-convention re-run changed.

Corrections to the round-4 log itself (EXPERIMENT_LOG, October 22) that this report applies: "no monitor has a significant gain; 2 have significant losses" becomes "after Holm only Luna's loss is significant"; "a property of weak-to-mid monitors" is dropped (read-out effect); the pooled MOLE interval uses the joint account bootstrap; "Luna's insider signal is mostly account-level" is dropped; the leave-apps-out realized-FPR range is 0.049–0.109, not 0.014–0.117; "today's score is best or tied everywhere" becomes 9 of 11; "max is as good as any forgetting rule" holds only as non-significance at $\mathrm{ARL}_0 = 100$.[53]

14 Errata relative to the October 21 version

Table 15 lists every statement of the October 21 version of this report that the audits changed, with the corrected statement and its source. Statements not listed (e.g. the round-1 TPR tables and the k-grid) are unchanged point estimates.

[53] Source: `results/audit3/corrections.csv`.

October 22 statement	Corrected statement	Source
GPT-4o top-3 − max +0.146 [+0.025, +0.226]	+0.134 [+0.024, +0.217] (conformal rule); holds	fixes2
gpt-4.1-mini PR gain +0.145 [−0.001, +0.228], borderline	+0.145 [+0.007, +0.230] (conformal); app-cluster CI covers 0; exists only with the logprob read-out	fixes2; audit 3
"Deployed (no-reset) monitoring needs a forgetting accumulator" (CUSUM 0.71 vs 0.58)	Point estimate reproduces; +0.132 [−0.086, +0.296] with CIs; pooled over 9 monitors no forgetting rule beats max	r4_cs; audit 3
Nested chooser +0.001 [−0.014, +0.013]	+0.001 [−0.006, +0.009] (conformal; both arms now at 1.01% realized FPR instead of 2.04%)	fixes2
MOLE pooled single-day 3rd-max −0.066 [−0.156, +0.023]	−0.064 [−0.151, +0.024] (conformal)	fixes2
Cross-session accumulation is "the one positive regime"	Not general: pooled over nine monitors −0.066 [−0.147, +0.015]	r4_cs
Thresholds: randomized plug-in primary	Conformal randomized rule; plug-in cells labelled	audit 3; fixes2

Table 14: Changes relative to the October 22 version. "fixes2" = `results/fixes2/headline_threshold_rerun.csv`; "audit 3" = `results/audit3/corrections.csv`.

October 21 statement	Corrected statement	Source (issue)
Top-3 sum vs max on history-free PR monitor +0.15 [0.12, 0.17]	+0.146 [+0.025, +0.226]; holds with a wider CI (corrected bootstrap); +0.134 [+0.024, +0.217] under the conformal rule	fixes C2
"The gain disappears once the per-PR monitor sees prior diffs"	True for GPT-4o only; gpt-4.1-mini gains more with history; Qwen, Gemini n.s.	fixes C2
"First matched-budget evidence that cross-session accumulation beats max"	Monitor-specific: GPT-4o holds, gpt-4.1-mini borderline, Gemini/Qwen none; needs a forgetting statistic under continuous monitoring	fixes C2, m9; r3_t1
Accumulation predicted by honest persistence ($\rho = -0.67$) and repeat-spike informativeness (+0.47)	Descriptive only; out-of-sample +0.30, family CI [−0.03, +0.45]; −0.56 [−0.69, −0.05]	fixes M7
Conclusion 3: choose the rule from measurable stream properties	Dropped: neither the mechanism rule nor the empirical chooser beats always-max out of sample	fixes M6, M7
MB CUSUM DiD +0.02 at $\alpha = 0.01$	+0.024 [−0.050, +0.072], n.s.	fixes C2
MB DiD negative at $\alpha = 0.05$	Instruction artifact; near zero with matched prompts	fixes M4
Full history makes the monitor re-flag attacks (S1: 60% vs 39%)	Matched prompts: S1 +0.08 [−0.08, +0.24], all +0.03 [−0.03, +0.09]	fixes M4
Max TPR none 0.322 > full 0.226 ($\alpha = 0.05$)	Reverses with matched prompts: none − full −0.091 [−0.186, +0.038]	fixes M4
"About 17–20% TPR at 1% trajectory FPR"	opus-4-5 attacks (like-for-like): 0.121 (full), 0.117 (none)	fixes M9
"Realized FPRs within about ±0.01 of target"	At $\alpha = 0.01$ realized/target 0.77–2.30 (mean 1.28)	fixes m1
Weak monitor: top-3 sum 0.082 vs max 0.056	Holds: +0.026 [+0.001, +0.058]	fixes C2
MRT length-aware max 0.21 → 0.58	Holds: +0.365 [+0.127, +0.445]; attacker can pad length	fixes m7
Max alarms at ideal flag step (median delay 0)	Partly by construction (onset is an LLM judgment)	fixes m12
Next step: build an adaptive rule chooser	Built in round 2; fails out of sample	fixes M6
Spend $109.56	Round 1 only; rounds 2–3 and audit listed in the summary	log

Table 15: Errata. "fixes" = `results/fixes/corrections.csv`; issue ids from `results/audit/issues.csv`.